Novosibirsk State University

*THE REAL SCIENCE

TELECOMMUNICATION CUSTOMERS CHURN PREDICTION USING HYBRID NEURAL NETWORK AND INTERPRETABLE ML(XAI) METHODS.

APPLIED MATHEMATICS AND INFORMATICS — ARTIFICIAL INTELLIGENCE AND BIG DATA ANALYSIS.

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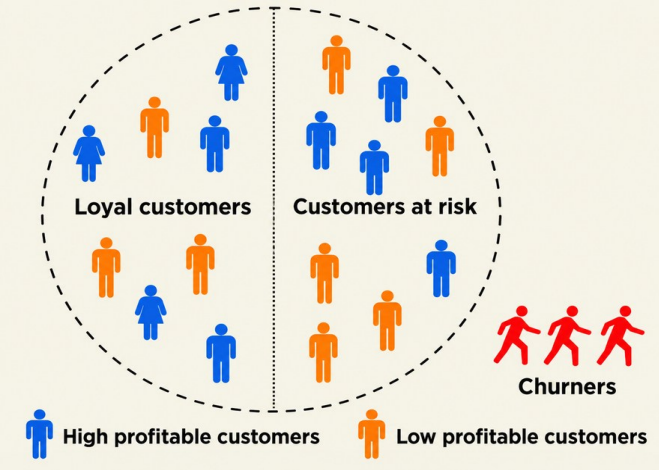
RELEVANCE

- **Relevance:**

- In a hypercompetitive telecom environment:

→ High risk of customer churn.

→ Acquisition costs > retention costs.





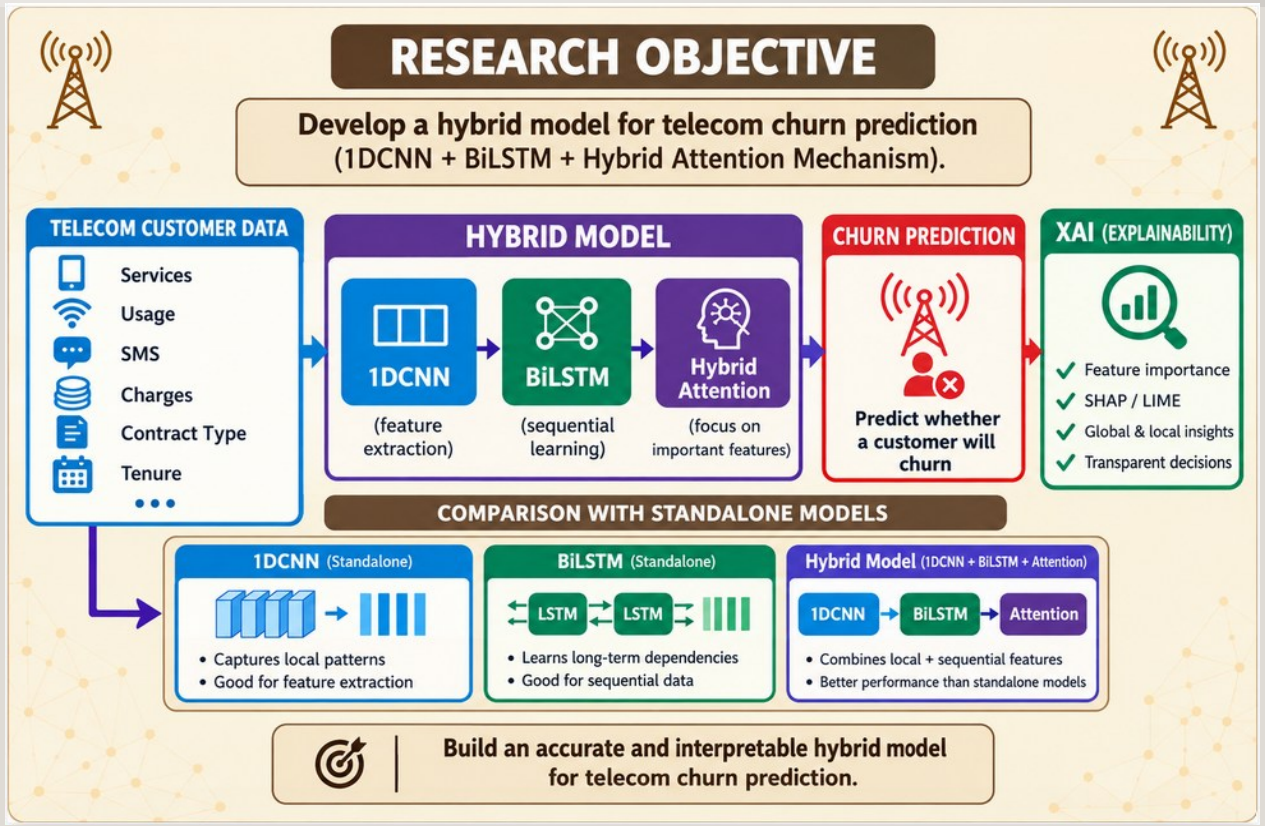
OBJECTIVE OF THE RESEARCH

- **Research objective:**

→ Develop a hybrid model:
(1DCNN + BiLSTM +
hybrid attention mechanism).

→ Comparison with standalone models:
(1D CNN, BiLSTM).

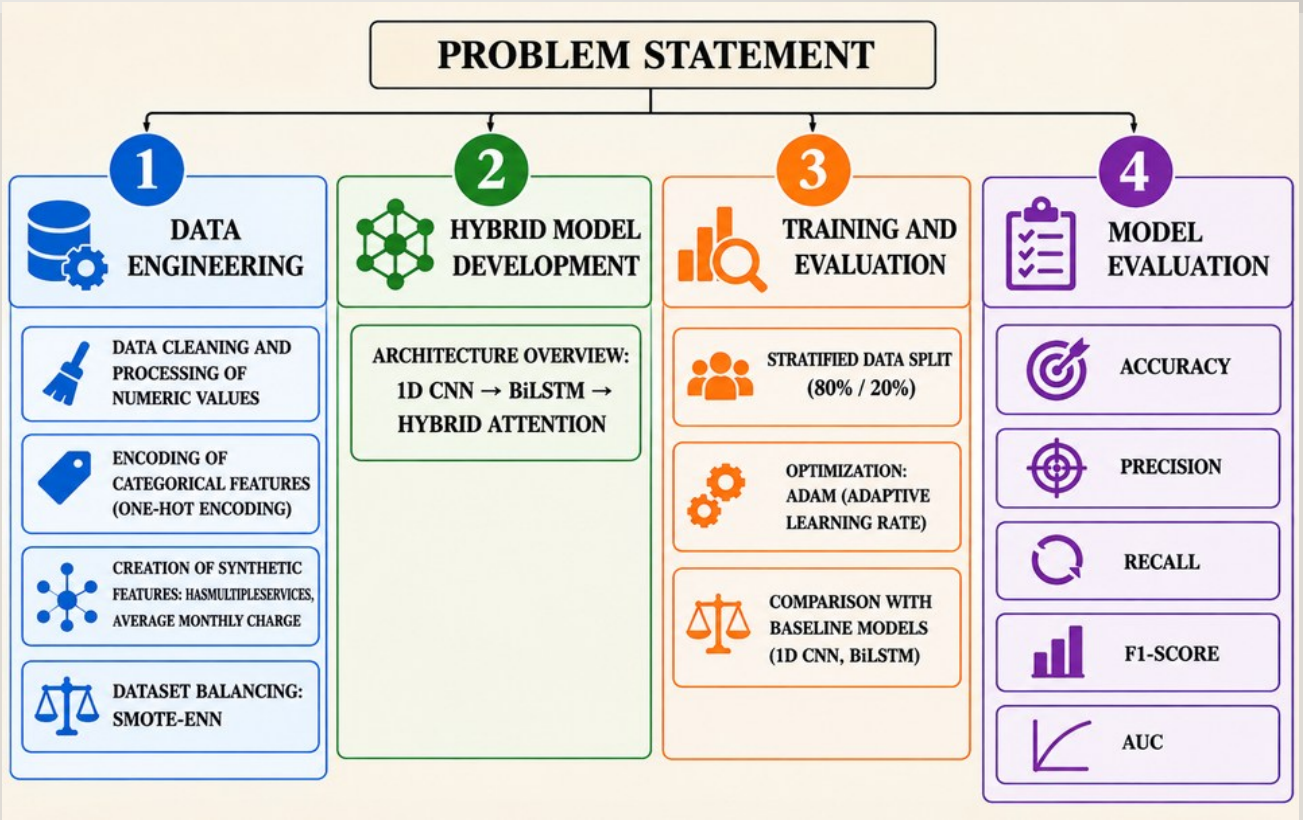
→ Explain the hybrid model prediction.



PROBLEM STATEMENT

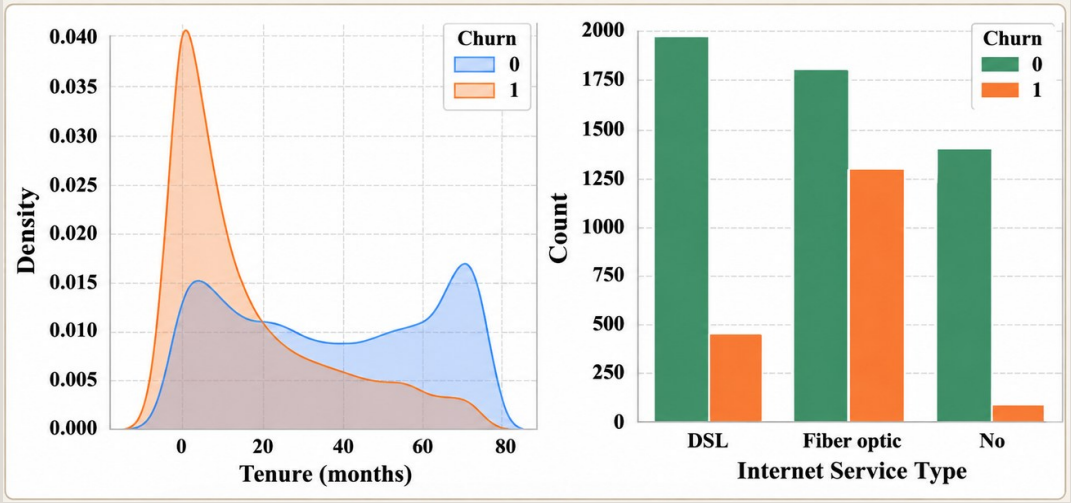
- **To achieve the stated goal, the following parts of tasks were implemented:**

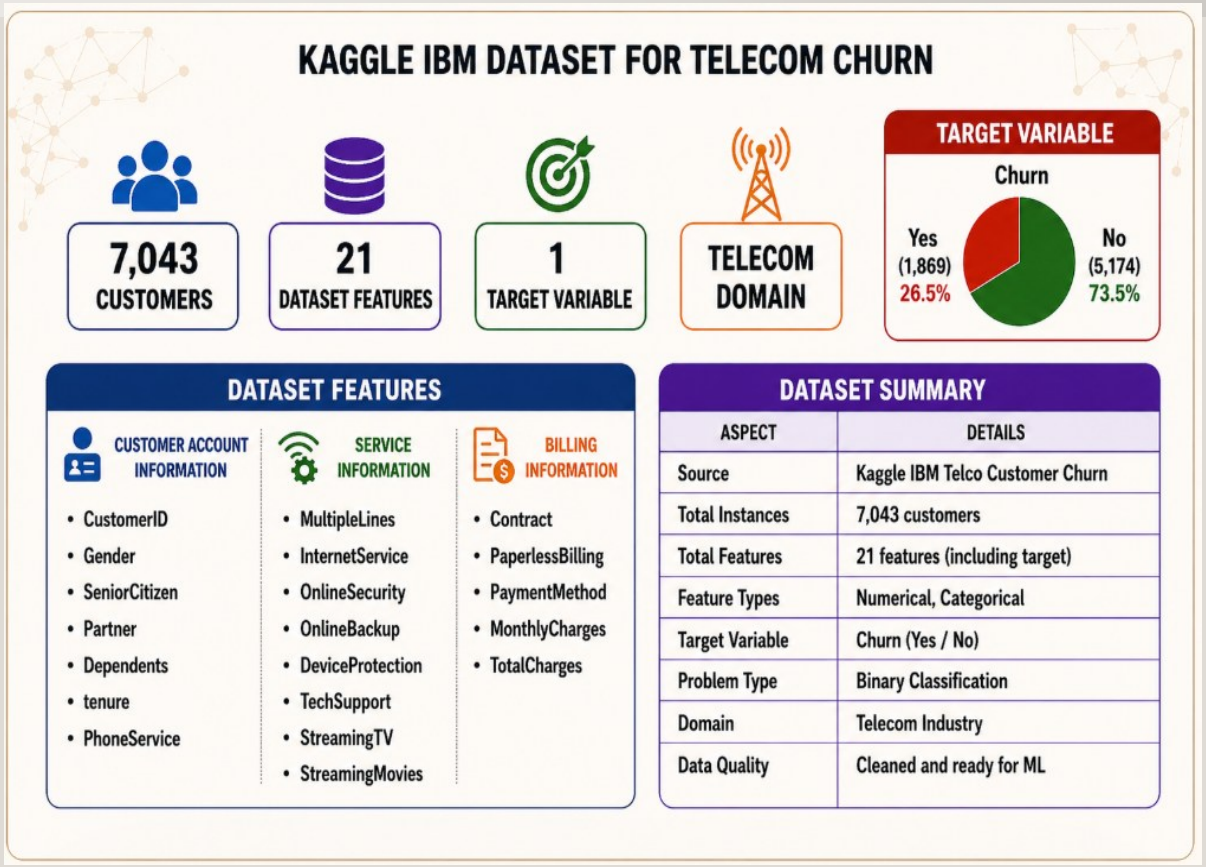
1. Data Engineering.
2. Development of the model architecture.
3. Training of the hybrid model.
4. Evaluation of the hybrid model.
5. Explanation of the hybrid model.
6. Finding of significant features.



DATA PREPARATION AND FEATURE ENGINEERING

- Overview of the dataset:
  (IBM Telco, 7043 customers, 21 features).
- Customenrs: (0–5 months) — new(churn).
  > 65 months — loyal (not churn).
- Fiber-optic internet users are more prone to churn.





DATA PREPARATION AND FEATURE ENGINEERING (CONTINUED…)

- **Filling null-values, OHE(Cat.), Scaling(Num.).**
- **Synthetic(engineered) features:**

→ HasMultipleServices:

$$HasMultipleServices = \sum_{j=1}^{ServiceCount} (Service_j = Active)$$

→ AverageMonthlyCharge:

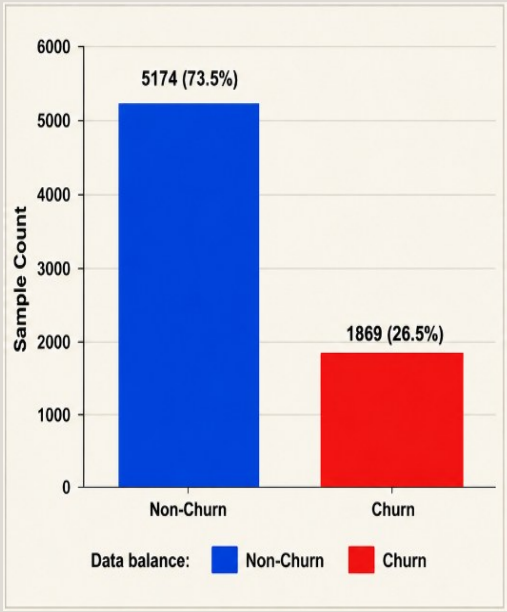
$$AverageMonthlyCharge = \frac{TotalCharges}{tenure + 1}$$

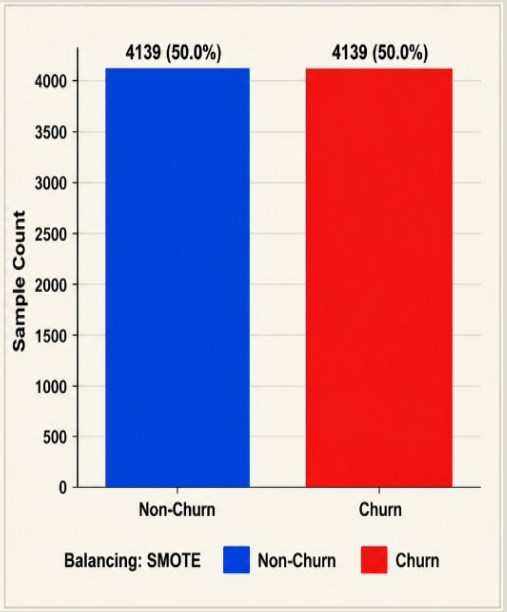
- **Dataset balancing**:

→ SMOTE

(73.5%→50.0% Non-Churn ,

26.5%→ 50.0% Churn).



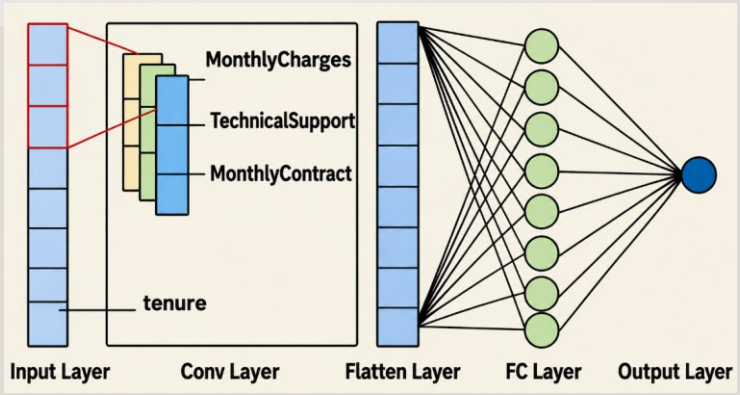


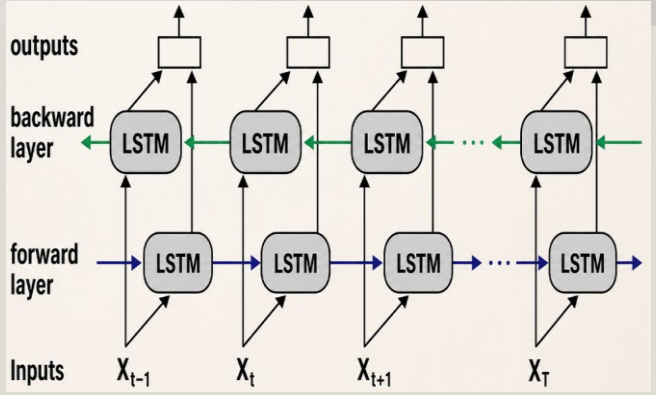
CustomerID	Contract	InternetService	PaymentMethod	Monthly Charges
A001	month-to-month	Fiber	Electronic check	$50.000
A002	one-year	DSL	Mailed check	$75.273
A003	two-years	No	Bank transfer (automatic)	$56.355

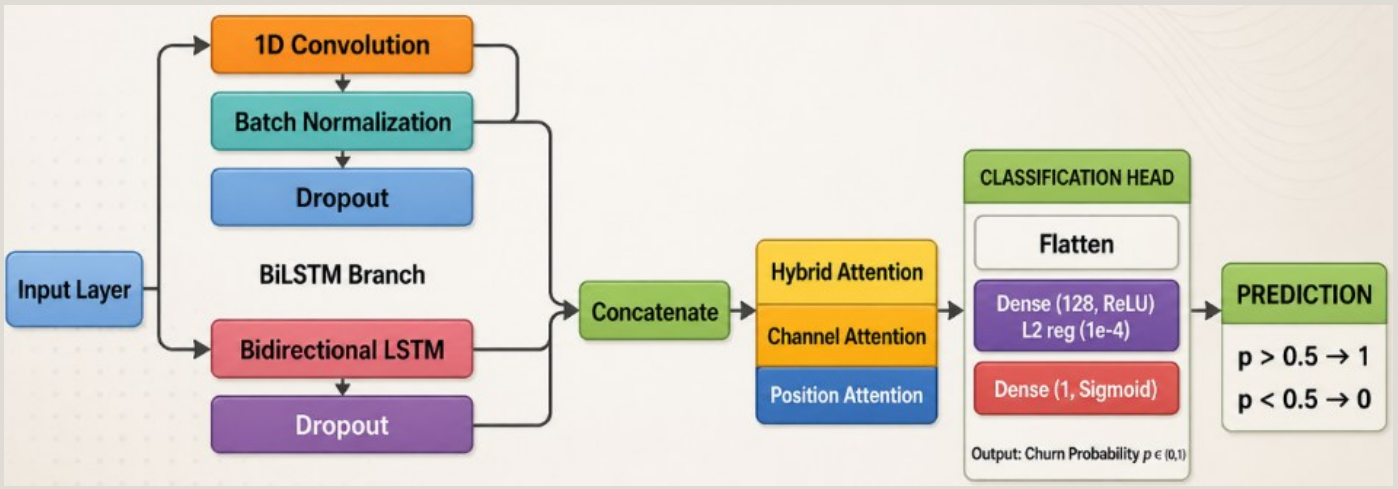
Encoding	Normalizing Numerical Features			Feature Matrix			
Categorical Features	Tenure	Monthly Charges	Total Charges	f1	f2	f3	...
One-Hot Encoding	19.43	29.35	3.18	29.45	64.45	63	...
	69.07	56.95	6.0	23.07	53.35	82	...
	49.221	60.23	0.0	49.82	53.35	78	...
	Total/Avg	63.228	29.25	53.55	53.55	55	...

IMPLEMENTED DEEP LEARNING ARCHITECTURES

- **1D CNN** → detection of local patterns along the feature axis.
- **BiLSTM** → Capture sequential patterns from both past and future contexts.
- **Proposed hybrid model:** 1D CNN + BiLSTM + hybrid attention (channel + positional).





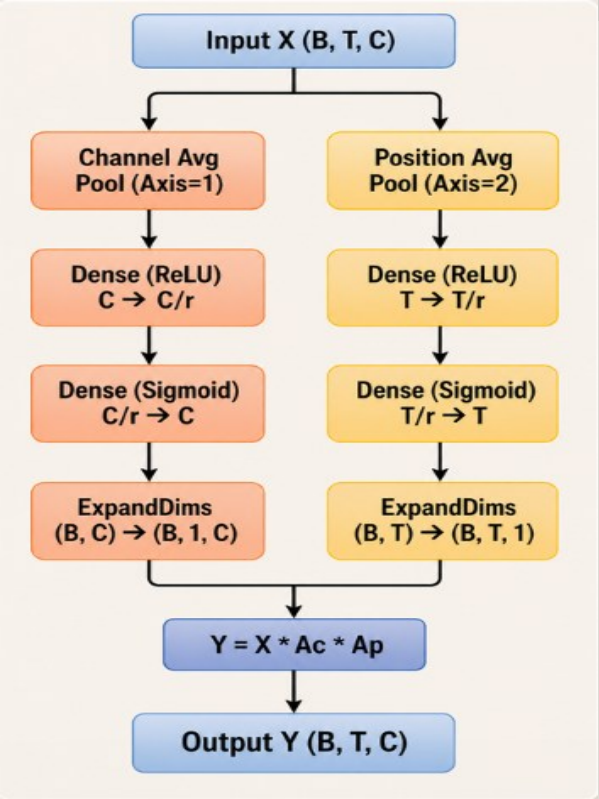


HYBRID ATTENTION MECHANISM (KEY NOVELTY)

- Input data: concatenated 1D CNN - BiLSTM features.
- Components → (channel attention + positional attention).
- Channel attention → which features are important?
- Positional attention → where are they important?
- Dynamically extract and weight significant features.
- Simplified mathematical formula: $Y = X * A_c * A_p$

  Where ***X***— combined feature map, ***Y*** — refined feature map,

  Ac — channel attention weights, ***Ap*** — position attention weights.
- Output data: recalibrated features.



HYPERPARAMETER CONFIGURATIONS OF THE EVALUATED MODELS.

Model	Category	Parameter	Specification / Value
CNN	Architecture	Layers & Activation	1× Conv1D (64 filters, kernel size 3, ReLU) + 1× BatchNormalization + 1× Conv1D (128 filters, kernel size 3, ReLU) + BatchNormalization
		Pooling	MaxPooling1D (pool size = 2, Dropout: 0.2) + GlobalAveragePooling1D (Dropout: 0.3)
	Training	Optimization & loss	Adam, Binary Crossentropy
BiLSTM	Architecture	Recurrent Layers	2× Stacked Bidirectional LSTM (1st layer: 64 units, 2nd layer: 32 units)
		Regularization	L2 = 0.001, Dropout: 0.3 (applied after each BiLSTM layer)
	Training	Optimization	Adam, Binary Crossentropy
CNN-BiLSTM-HybridAttention	Architecture	CNN Branch	64 filters, kernel size 3 (padding="same"), ReLU, L2 = 1e-4, Dropout: 0.3, Batch Normalization
		LSTM Branch	Bidirectional LSTM (64 units, return_sequences=True), L2 = 1e-4, Dropout: 0.3
		Attention Module	HybridAttention1D (Reduction ratio = 8, Dropout: 0.4, L2 = 1e-4)
		Classification Head	Flatten, Dense layer (128 units, ReLU, L2 = 1e-4, Dropout: 0.4), Output: Dense(1, Sigmoid)
	Training	Optimization	Adam (lr = 0.0003, weight decay = 1e-6), Focal Loss (γ=2.5, α=0.75)

RESULTS OF THE COMPUTATIONAL EXPERIMENT.

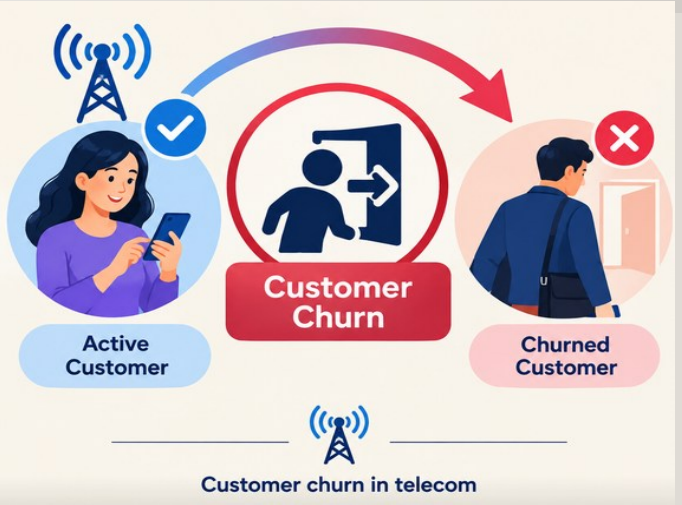
- **Performance metrics comparison:**

→ (Accuracy, Precision, Recall, F1-Score, AUC).

- **The hybrid model shows best results:**

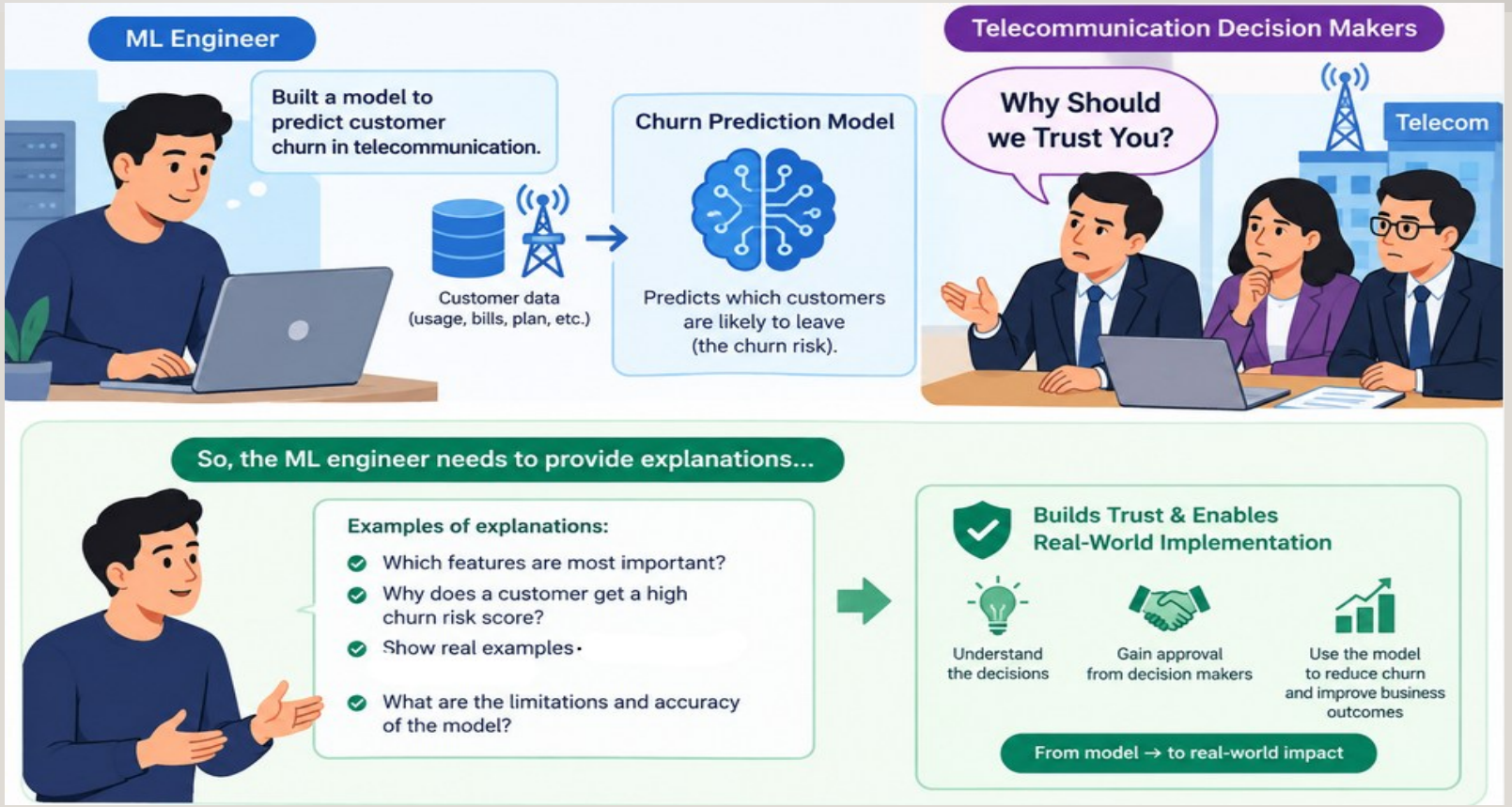
→ (Accuracy 82.04%, F1-Score 67.52% and AUC 0.86).

Model	Accuracy	Precision	Recall	F1-Score	AUC
CNN	77.36%	54.96%	80.16%	65.21%	0.86
BiLSTM	62.60%	41.07%	94.91%	57.33%	0.85
CNN-BiLSTM-HybridAttention	82.04%	64.78%	70.51%	67.52%	0.86



WHY WE SHOULD WE TRUST YOU?

Explain for us?



PLAN OF EXPLAINABILITY AFTER TRAINING (WHY?)

Integrate four explainable methods:

→ **SHAP**, Partial Dependence Plot (**PDP**),

Differential Accumulated Local Effect (**DALE**) and **LIME**.

- Global Explainability:

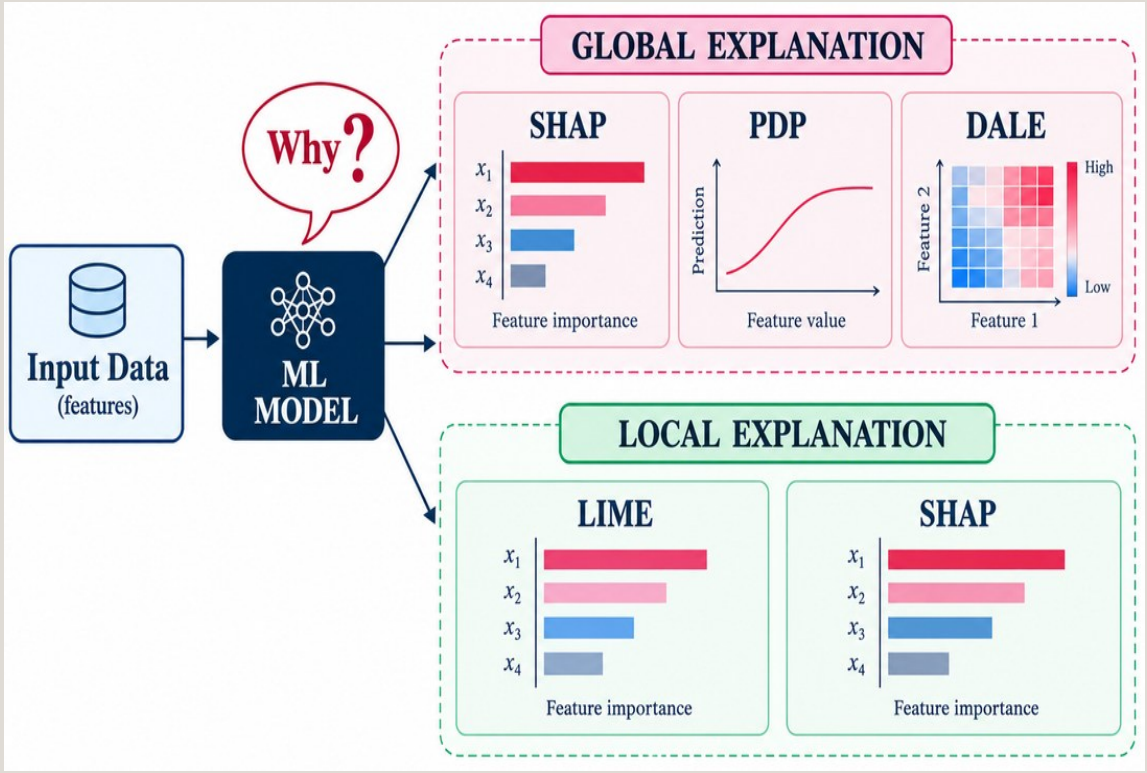
→ **SHAP**(Summary plot, and Interaction plot),

PDP and **DALE**.

- Local Explainability:

→ **SHAP**(waterfall plot, force plot, decision plot),

LIME(Bar chart, Decision plot).

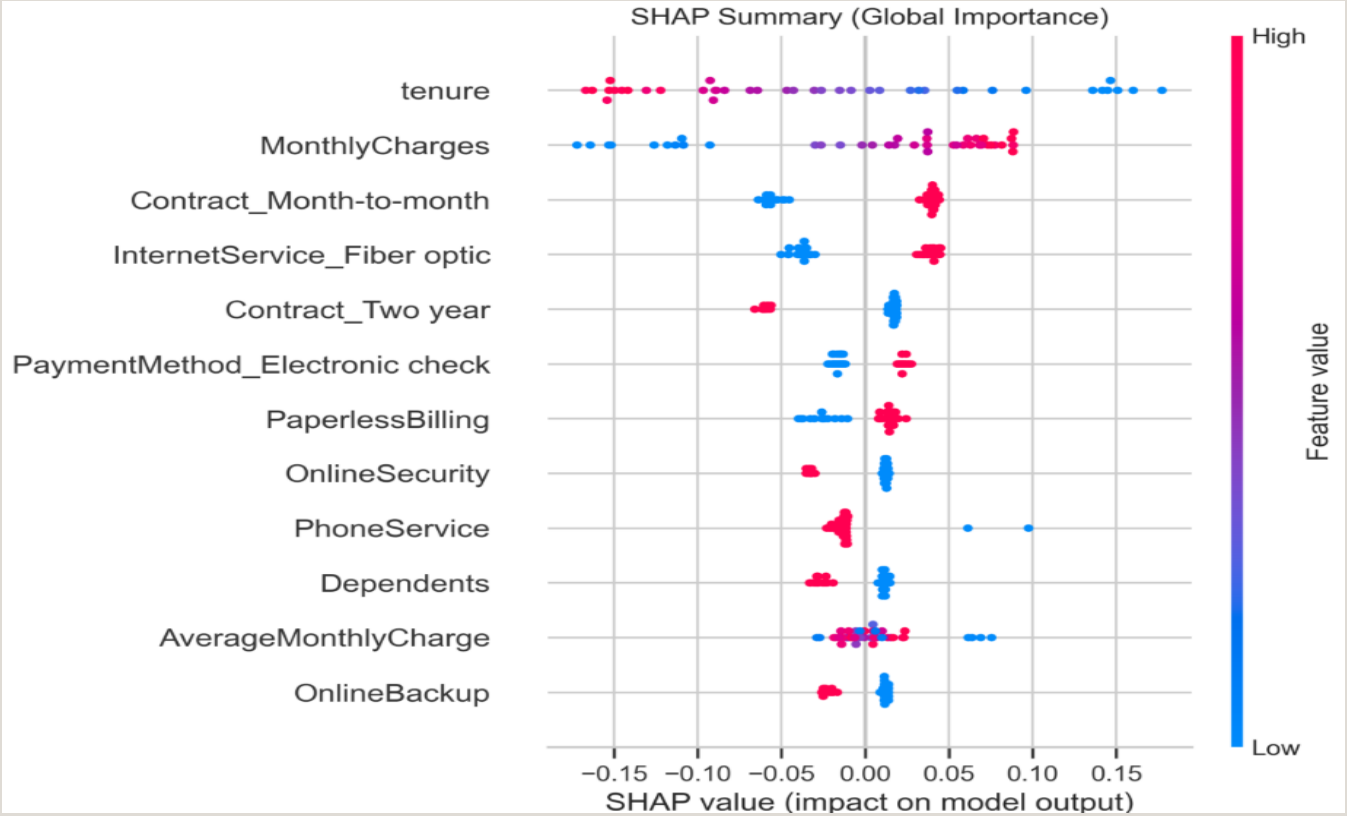


GLOBAL EXPLAINABILITY USING SHAP (SHAP SUMMARY PLOT)

SHAP Summary plot:

→ Explains the entire model's behavior across all customers.

- Red flagged colors:
  → Churn (Exit). High.
- Blue flagged colors:
  → Not-churn (Stay). Low.
- Top four important features:
  → Tenure, MonthlyCharges, Month-to-Month contract, Fiber-Optic Internet service.

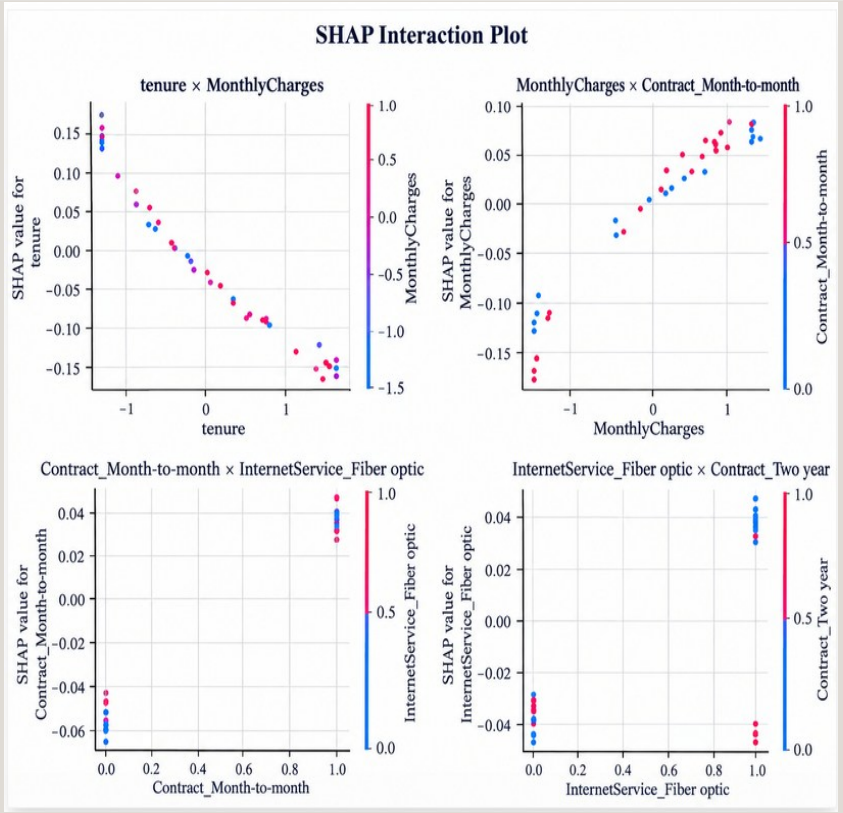


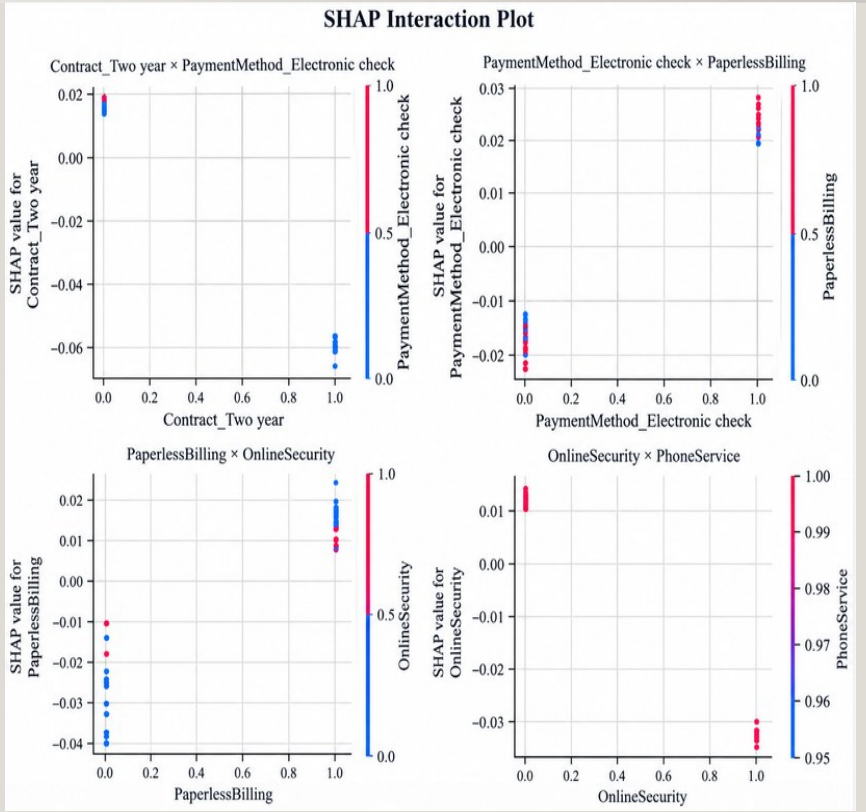
GLOBAL EXPLAINABILITY USING SHAP (SHAP INTERACTION PLOT)

SHAP Interaction Plot:

- The effect of a single feature on the prediction, conditioned on the value of a second feature.

- Shorter customer *tenure* strongly increases churn risk, and this risk is slightly amplified for those with higher monthly charges.

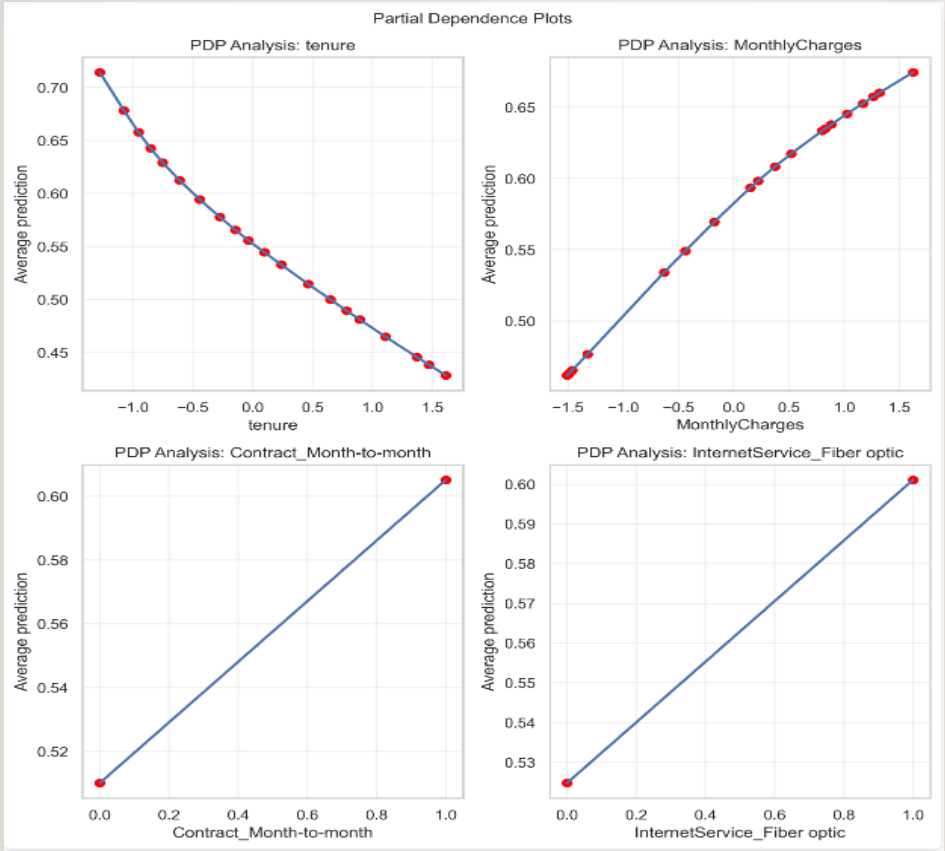


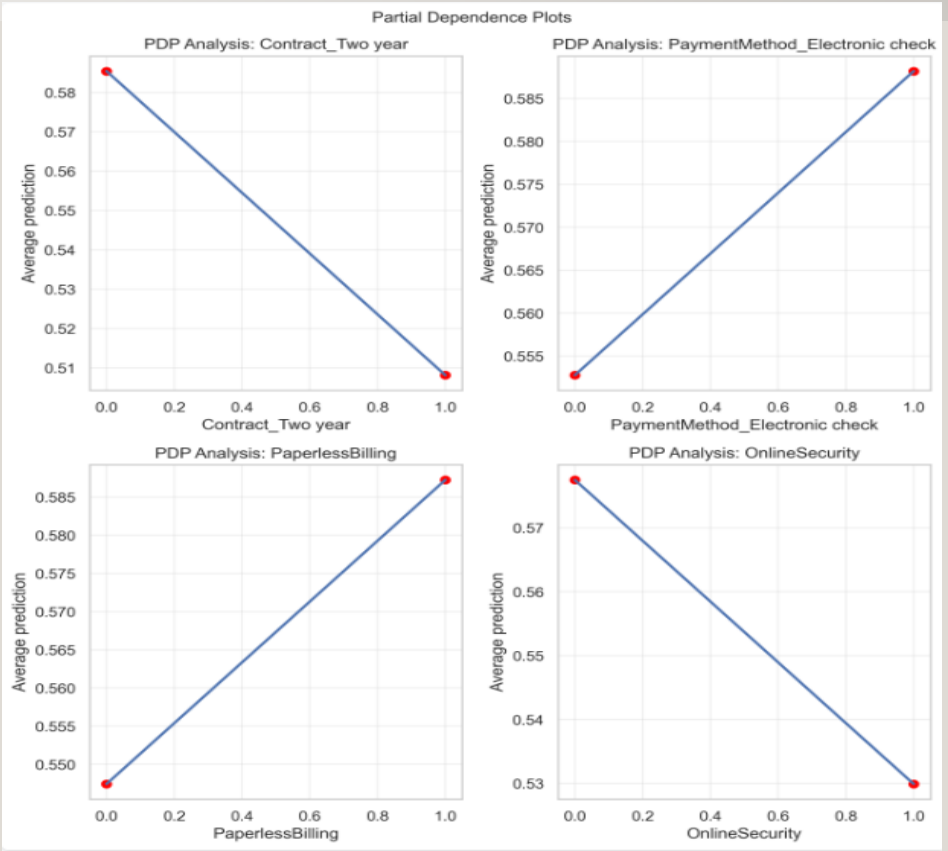


GLOBAL EXPLAINABILITY USING PDP(PARTIAL DEPENDENCE PLOT)

Partial Dependence Plot:

- Average prediction of a single feature over the entire customer base.





CONCLUSION AND OUR NEXT WORK

- **Conclusion:**

→ The hybrid model outperforms standalone 1D CNN and BiLSTM models.

→ Global key features: *tenure, monthly charges, month-to-month contract, and fiber optic internet service*.

- **Next work:**

→ *Global Explainability*:

→ Differential Accumulated Local Effect (DALE)

→ *Local Explainability*:

→ **SHAP**(waterfall plot, force plot, decision plot, feature importance plot), **LIME**(Bar chart, Decision plot).

THE END!

Thank you !!!

